Week 1 Assignment: Python Fundamentals for Data Analytics

Submission Package

This report contains all required tasks, brief explanations, and complete Python source code for each program.

Objective

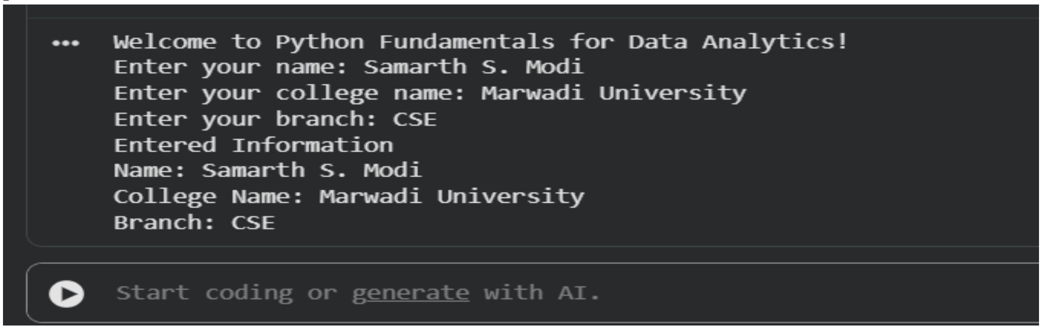
To build a strong foundation in Python programming for data analytics by practicing variables, data types, control flow, functions, collections, file handling, and a mini project.

Task 1: Python Basics

Program to print a welcome message, take name, college name, and branch as input, and display them in formatted output.

Source Code:

```
print("Welcome to Python Fundamentals for Data Analytics!")
name = input("Enter your name: ")
college = input("Enter your college name: ")
branch = input("Enter your branch: ")
print("
Entered Information")
print(f"Name: {name}")
print(f"College Name: {college}")
print(f"Branch: {branch}")
```



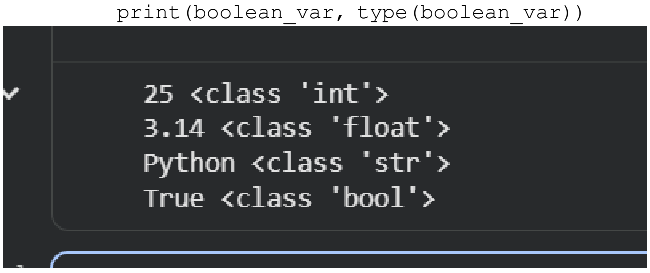
Task 2: Variables & Data Types

Program to declare integer, float, string, and boolean variables and print each value with its data type using type().

Source Code:

```
integer_var = 25
float_var = 3.14
string_var = "Python"
boolean_var = True

print(integer_var, type(integer_var))
print(float_var, type(float_var))
print(string_var, type(string_var))
print(boolean_var, type(boolean_var))
```



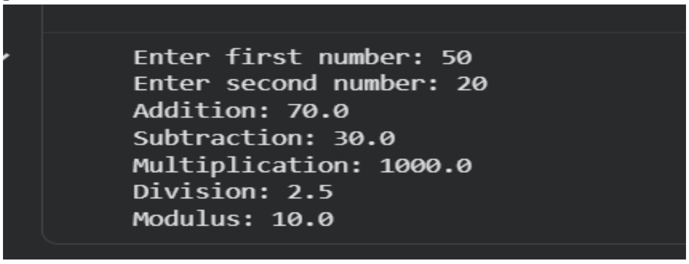
Task 3: Operators

Calculator program that performs addition, subtraction, multiplication, division, and modulus using two user-entered numbers.

Source Code:

```
num1 = float(input("Enter first number: "))
num2 = float(input("Enter second number: "))

print("Addition:", num1 + num2)
print("Subtraction:", num1 - num2)
print("Multiplication:", num1 * num2)
print("Division:", num1 / num2)
print("Modulus:", num1 % num2)
```



```
Enter first number: 50
Enter second number: 20
Addition: 70.0
Subtraction: 30.0
Multiplication: 1000.0
Division: 2.5
Modulus: 10.0
```

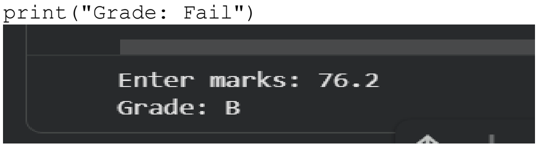
Task 4: Conditional Statements

Program that accepts marks and displays a grade using if, elif, and else based on the given marks range.

Source Code:

```
marks = float(input("Enter marks: "))

if marks >= 90:
    print("Grade: A")
elif marks >= 75:
      print("Grade: B")
elif marks >= 60:
    print("Grade: C")
else:
    print("Grade: Fail")
```



```
Enter marks: 76.2
Grade: B
```

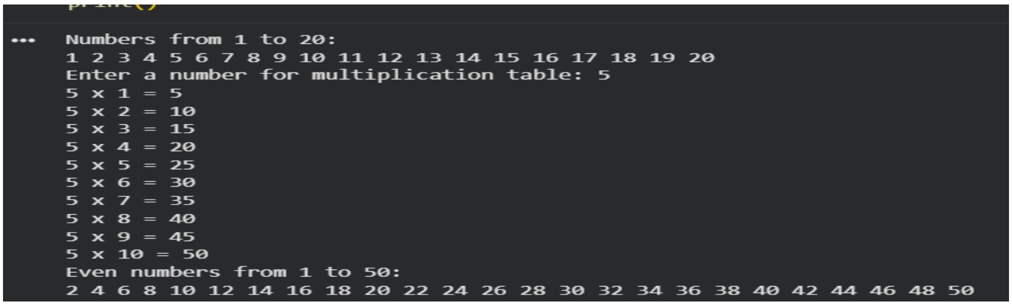
Task 5: Loops

Programs to print numbers from 1 to 20 using a for loop, print a multiplication table, and print even numbers from 1 to 50 using a while loop.

Source Code:

```
print("Numbers from 1 to 20:")
for i in range(1, 21):
    print(i, end=" ")

number = int(input("
Enter a number for multiplication table: "))
for i in range(1, 11):
    print(f"{number} x {i} = {number * i}")
print("Even numbers from 1 to 50:")
i = 1
while i <= 50:
    if i % 2 == 0:
        print(i, end=" ")
    i += 1
```



```
Numbers from 1 to 20:
1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17 18 19 20
Enter a number for multiplication table: 5
5 x 1 = 5
5 x 2 = 10
5 x 3 = 15
5 x 4 = 20
5 x 5 = 25
5 x 6 = 30
5 x 7 = 35
5 x 8 = 40
5 x 9 = 45
5 x 10 = 50
Even numbers from 1 to 50:
2 4 6 8 10 12 14 16 18 20 22 24 26 28 30 32 34 36 38 40 42 44 46 48 50
```

Task 6: Functions

Two user-defined functions are created: one for square of a number and one for average of three numbers.

Source Code:

```
def square(n):
    return n * n

def average(a, b, c):
    return (a + b + c) / 3

num = float(input("Enter a number for square: "))
a = float(input("Enter first number: "))
b = float(input("Enter second number: "))
c = float(input("Enter third number: "))

print("Square:", square(num))
print("Average:", average(a, b, c))
```

```
Enter a number for square: 9
Enter first number: 5
Enter second number: 6
Enter third number: 7
Square: 81.0
Average: 6.0
```

Task 7: Strings & Collections

Demonstrates string methods, list operations, tuple indexing, dictionary storage, and set operations.

Source Code:

```
text = "Python Data Analytics"
print(text.upper())
print(text.lower())
print(text.replace("Python", "Advanced Python"))
print(text.find("Data"))

items = ["Apple", "Banana", "Cherry"]
items.append("Orange")
items.remove("Banana")
items.sort()
print(items)

tuple_data = (10, 20, 30, 40)
print(tuple_data)
print(tuple_data[2])

student = {"name": "Aman", "age": 20, "branch": "CSE"}
print(student)

s = {1, 2, 3}
s.add(4)
s.remove(2)

print(s)
```

```
PYTHON DATA ANALYTICS
python data analytics
Advanced Python Data Analytics
7
['Apple', 'Cherry', 'Orange']
(10, 20, 30, 40)
30
{'name': 'Aman', 'age': 20, 'branch': 'CSE'}
{1, 3, 4}
```

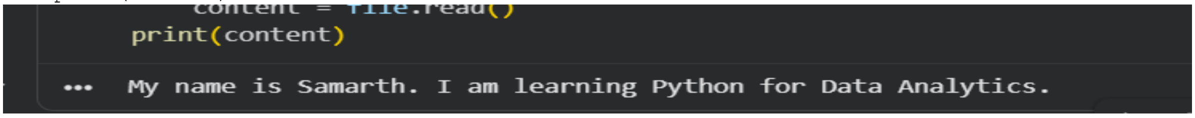
Task 8: Basic File Handling

Program that creates a text file, writes an introduction, and reads the file contents back.

Source Code:

```
with open("introduction.txt", "w") as file:
    file.write("My name is Rahul. I am learning Python for Data Analytics.")

with open("introduction.txt", "r") as file:
    content = file.read()
    print(content)
```



Task 9: Mini Python Project

Student Record Management System using a list of dictionaries. It supports adding, displaying, searching, and deleting student records.

Source Code:

```
students = []

def add_student():
    name = input("Enter student name: ")
    roll = input("Enter roll number: ")
    branch = input("Enter branch: ")
    marks = input("Enter marks: ")
    students.append({"name": name, "roll": roll, "branch": branch, "marks": marks})
    print("Student added successfully.")

def display_students():
    if not students:
        print("No records found.")
        return
    for i, student in enumerate(students, start=1):
        print(f"{i}. {student}")
def search_student():
    name = input("Enter name to search: ")
    found = False
    for student in students:
        if student["name"].lower() == name.lower():
            print("Student found:", student)
            found = True
            break
    if not found:
        print("Student not found.")

def delete_student():
    name = input("Enter name to delete: ")
    for student in students:
        if student["name"].lower() == name.lower():
            students.remove(student)
            print("Student record deleted.")
            return
    print("Student not found.")

while True:
    print("
1. Add Student")
    print("2. Display Students")
    print("3. Search Student")
    print("4. Delete Student")
    print("5. Exit")
    choice = input("Enter choice: ")
    if choice == "1":
        add_student()
    elif choice == "2":
        display_students()
    elif choice == "3":
        search_student()
    elif choice == "4":
        delete_student()
    elif choice == "5":
        break
    else:
        print("Invalid choice")
```

```
...
1. Add Student
2. Display Students
3. Search Student
4. Delete Student
5. Exit
Enter choice: 1
Enter student name: Samarth Modi
Enter roll number: 0270
Enter branch: CSE
Enter marks: 76.2
Student added successfully.

1. Add Student
2. Display Students
3. Search Student
4. Delete Student
5. Exit
Enter choice: 5
```

Conclusion

This assignment covers the foundational concepts of Python required for data analytics and provides hands-on practice through small programs and a mini project.

Google Drive Link: https://drive.google.com/drive/folders/1p3o60ule-BRJsyhxr0KQj-1C57zXnpgN?usp=sharing